

The Fundamental Theorem of Calculus (5.3)

Zero's arrow flies in a straight line with velocity $v(t)$, during the time interval $[a, b]$. What is its position at time T , where $a \leq T \leq b$?

If $v(t)$ is constant on some interval of length Δt , then it travels $v(t)\Delta t$ during that time.

Assume $s(a) = 0$.

Subdivide $[a, T]$ into smaller intervals of length Δt . Choose some t_i^* from each small interval, and assume v takes the constant value $v(t_i^*)$ on that interval.

The total distance the arrow travels is approximated by

$$\sum_{i=1}^n v(t_i^*) \Delta t$$

Since we assumed $s(a) = 0$,

$$s(T) \approx \sum_{i=1}^n v(t_i^*) \Delta t$$

$$s(T) = \int_a^T v(t) dt$$

We also know $s'(T) = v(T)$

Fundamental theorem of calculus, part 1

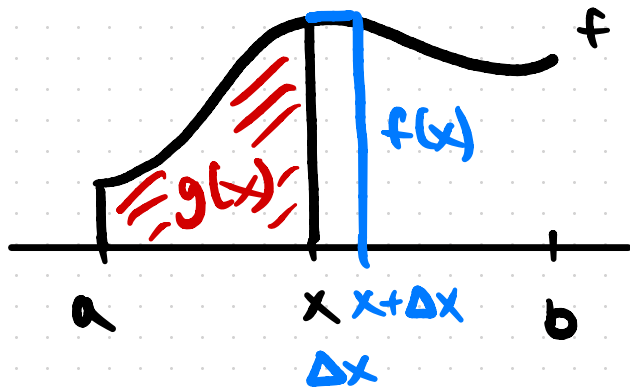
Let f be a continuous function defined on $[a, b]$,
and let

$$g(x) = \int_a^x f(t) dt$$

✓ t is local to the integral
"dummy variable"

for $a \leq x \leq b$. Then $g'(x) = f(x)$ for all $a < x < b$.

The function $g(x)$ measures the signed area under the graph of f from a to x .



$g'(x)$ is the rate of change
of g at x

Say x changes by a small amount Δx
Then $g(x)$ changes by approximately
the blue rectangle, which has area $f(x)\Delta x$.

$$g'(x) \approx \frac{\Delta g(x)}{\Delta x} = \frac{f(x)\Delta x}{\Delta x} = f(x)$$

Example: Let

$$g(x) = \int_3^x (t^4 + t^3) dt$$

What is $g'(x)$?

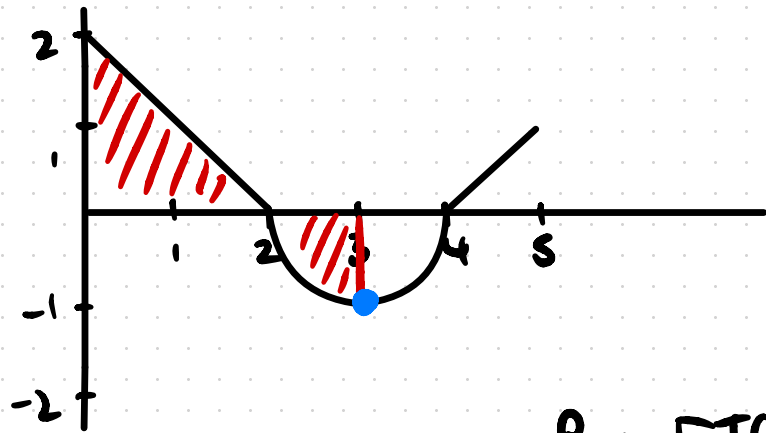
By FTC


$$g'(x) = x^4 + x^3$$

Example: Let

$$g(x) = \int_0^x f(t) dt$$

where f is the function.



$$g(3) = \int_0^3 f(t) dt$$
$$= 2 - \frac{\pi}{4}$$

By FTC

$$g'(3) = f(3)$$
$$= -1$$

Example: Let

$$g(x) = \int_2^{x^2} t^3 dt$$

What is $g'(x)$?

Let $u = x^2$

$$g(x) = \int_2^u t^3 dt$$

$$g'(x) = \frac{d}{dx} \int_2^u t^3 dt$$

$$= \left(\frac{d}{du} \int_2^u t^3 dt \right) \cdot \frac{du}{dx}$$

$$= u^3 \cdot 2x$$

$$= (x^2)^3 \cdot 2x$$

$$= 2x^7$$

Example: Let

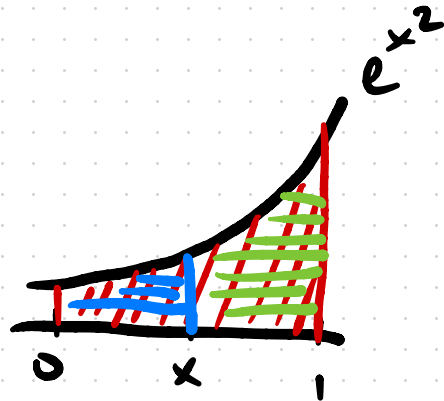
$$g(x) = \int_x^1 e^{t^2} dt$$

What is $g'(x)$?

Method 1:

$$g(x) = \underbrace{\int_0^1 e^{t^2} dt}_{\text{constant}} - \underbrace{\int_0^x e^{t^2} dt}$$

$$g'(x) = \underbrace{0}_{\text{derivative of a constant}} - e^{x^2} = -e^{x^2}$$



Method 2:

We set the notation that

$$\int_x^1 e^{t^2} dt = - \int_1^x e^{t^2} dt$$

It turns out the FTC still holds using this notation.

$$\begin{aligned} g'(x) &= - \frac{d}{dx} \int_1^x e^{t^2} dt \\ &= -e^{x^2} \end{aligned}$$

Example: Let

$$g(t) = \int_{\sqrt{t}}^{2t} \sin x \, dx$$

What is $g'(t)$?

$$g(t) = \int_0^{2t} \sin x \, dx - \int_0^{\sqrt{t}} \sin x \, dx$$

$$g'(t) = \frac{d}{dt} \int_0^{2t} \sin x \, dx - \frac{d}{dt} \int_0^{\sqrt{t}} \sin x \, dx$$

$$- \left(\frac{d}{d(2t)} \int_0^{2t} \sin x dx \right) \cdot \frac{d(2t)}{dt}$$

$$- \left(\frac{d}{d(\sqrt{t})} \int_0^{\sqrt{t}} \sin x dx \right) \cdot \frac{d(\sqrt{t})}{dt}$$

$$= (\sin 2t) \cdot 2 - (\sin \sqrt{t}) \cdot \frac{1}{2\sqrt{t}}$$

$$= 2 \sin 2t - \frac{1}{2\sqrt{t}} \sin \sqrt{t}$$

Zero's arrow flies in a straight line with velocity $v(t)$, during the time interval $[a, b]$. What is its position at time T , where $a \leq T \leq b$?

If we assume $s(a) = 0$, then

$$s(T) = \int_a^T v(t) dt$$

If we don't assume $s(a) = 0$, then

$$s(T) = s(a) + \int_a^T v(t) dt$$

$$s(T) - s(a) = \int_a^T v(t) dt$$

In particular, if $T=b$

$$\underline{s(b) - s(a) = \int_a^b v(t) dt}$$

net change
in position
from a to b

This holds for any s whose derivative is v .

Fundamental Theorem of Calculus, part 2.

Let f be a continuous function defined on $[a, b]$.

Then

$$\int_a^b f(x) dx = F(b) - F(a)$$

where F is any antiderivative of f .

This gives us an easy way to evaluate some definite integrals.

Example: Evaluate

$$\int_{-1}^1 (1-x^2) dx$$

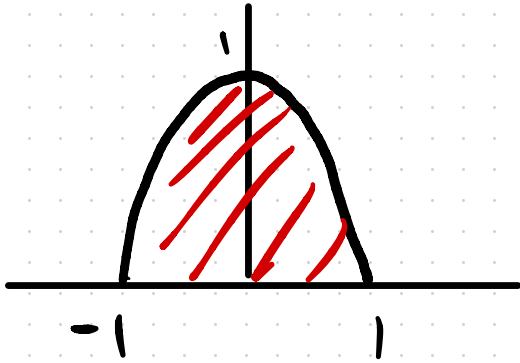
By the FTC,

$$\int_{-1}^1 (1-x^2) dx = F(1) - F(-1)$$

where F is any antiderivative of $1-x^2$.

Choose $F(x) = x - \frac{1}{3}x^3$.

$$\int_{-1}^1 (1-x^2) dx = \frac{2}{3} - \left(-\frac{2}{3}\right) \\ = \frac{4}{3}$$



$$\text{Area} = \frac{4}{3}$$

Notation: A common shorthand for
 $F(b) - F(a)$ is $F(x) \Big|_a^b$

$$\int_{-1}^1 (1-x^2) dx = \left(x - \frac{1}{3}x^3 \right) \Big|_{-1}^1$$

$$= \frac{2}{3} - \left(-\frac{2}{3} \right)$$

$$= \frac{4}{3}$$